COMPUTATIONAL ANALYSIS OF EMOTIONAL EXPRESSION, PSYCHOLOGICAL LANGUAGE, AND NARRATIVE EVOLUTION IN REDDIT COMMUNITIES

Final Research Report • Submission Edition

Scope: This report consolidates documented results from the project's saved analytical outputs and audit summaries. It preserves unresolved limitations instead of silently changing datasets or labels.

Executive Summary

This project examines Reddit discussions in gaming, science, todayilearned, and worldnews using sentiment/emotion outputs, linguistic features, topic modeling, cross-community statistical comparisons, and a temporal-topic audit. The Phase 8 merged analytical dataset contains 4,874 observations; the temporal topic dataset contains 4,026 records. Independent reproductions matched all 25 saved Kruskal–Wallis tests and all 150 pairwise comparisons.

VADER–transformer agreement varies by community, but agreement is not accuracy because neither method is treated as ground truth. Statistical significance is accompanied by small effect sizes in many tests. Topic labels are interpretive summaries, not ground-truth categories. The temporal dataset is severely imbalanced: 3,899 of 4,026 records (96.85%) fall in February 2016, so the current evidence does not support a strong claim of continuous month-to-month narrative evolution.

Objectives and workflow

- Characterize sentiment/emotion outputs and linguistic signals across four communities.
- Compare emotion-label patterns and model agreement.
- Explore candidate themes with topic modeling and audit labels.
- Assess whether temporal coverage supports narrative-evolution claims.
- Reproduce saved statistical calculations and document data-quality issues.

Data coverage

The master corpus contains 4,972 records; the processed NLP / Phase 8 corpus contains 4,874; the temporal topic-modeling corpus contains 4,026. The 848-record difference between the processed NLP and topic-modeling corpora remains unresolved. Separately, 98 master-corpus records are absent from the processed NLP corpus. These discrepancies are documented limitations.

Results

VADER–transformer agreement

Community	Agreement	Cohen's κ
gaming	70.84%	0.346
science	63.16%	0.191
todayilearned	68.32%	0.301
worldnews	67.78%	0.249

Agreement is highest in gaming and lowest in science. The kappa values suggest modest agreement beyond chance. Neither method is a ground-truth emotional labeler, so agreement cannot be interpreted as accuracy.

Association tests

VADER/transformer agreement by community: $\chi^2 = 11.646$, df = 3, p = 0.008699, Cramér's V = 0.04888. Emotion-label distribution: $\chi^2 = 87.778$, df = 18, p = 3.609 × 10■¹¹, Cramér's V = 0.07748. The p-values indicate evidence against the corresponding null hypotheses, but the Cramér's V values are small. Statistical significance does not imply a large practical effect.

Non-parametric comparisons

The saved analyses include 25 Kruskal–Wallis tests and 150 pairwise comparisons. Independent reproductions matched all 25 omnibus tests and all 150 pairwise comparisons, including statistics, raw p-values, and Holm adjustments. The saved results flag 22/25 omnibus tests and 91/150 pairwise comparisons as significant after Holm adjustment. Epsilon-squared values range approximately from 0.00055 to 0.0441, indicating generally small effects. These associations do not establish causation.

Topic modeling and label quality

The consolidated dictionary contains 40 model/community/topic groups. Five groups are marked do_not_label and should not be described as accepted labels. The quality audit found no missing labels, no short/generic label candidates, no exact duplicate labels, and two lexically similar label pairs. Lexical similarity is a screening heuristic, not proof of semantic identity. Batch approval is not equivalent to independent example-by-example human validation.

Temporal coverage

The temporal topic joins contain 4,026 unique IDs, with no duplicate IDs or missing dates, spanning 31 October 2014 to 17 February 2016. Only 10 of 17 calendar months have observations. February 2016 contains 3,899 records (96.85% of the temporal corpus). Because other observed months have small counts, the current dataset does not support a robust month-to-month trend or strong claim of continuous narrative evolution.

Validation, limitations, and conclusion

Computational validation

Structural checks passed for the saved temporal join tables and monthly prevalence calculations. The statistical audit checked p-value ranges, Holm adjustments, significance flags, and prevalence denominators. Independent reproductions matched all 25 Kruskal–Wallis tests and all 150 pairwise comparisons. These checks establish calculation consistency against saved inputs; they do not establish representativeness, measurement validity, or substantive correctness.

Limitations

1. **Corpus mismatch:** 4,874 Phase 8 records versus 4,026 topic-modeling records; the 848-ID difference remains unresolved.
2. **Master-to-NLP difference:** 98 master records are absent from the processed NLP corpus.
3. **Measurement:** VADER and transformer outputs are algorithmic estimates, not verified emotional states.
4. **Topic interpretation:** labels are model-dependent; five groups remain do_not_label and independent example-level validation is incomplete.
5. **Temporal imbalance:** 96.85% of topic records fall in February 2016.
6. **Inference:** statistical associations do not establish causality.
7. **Generalizability:** findings apply to the collected corpus and should not automatically be generalized to all Reddit users.

Conclusion

The project demonstrates an integrated computational workflow for comparing sentiment/emotion outputs, linguistic signals, and candidate topic structures across four Reddit communities. Its strongest contribution is a reproducible analysis accompanied by explicit statistical and data-quality audits. The results support cautious descriptive comparisons, but model agreement is imperfect, effect sizes are generally small, topic labels remain interpretive, and temporal coverage is highly imbalanced. Accordingly, the project does not claim causal effects or robust continuous narrative evolution.

Reproducibility note

This report summarizes existing saved outputs and audit summaries. It does not claim that the corpus discrepancy has been repaired, that every topic label has been independently validated, or that every raw input has been independently inspected. Consult the included CSV tables and audit reports for detailed results.